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Assignment 4

Q1 a) $\vec{a} = (2, 2, 1)$ $\vec{b} = (1, 0, -2)$

$$\text{proj}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{\|\vec{b}\|^2} \vec{b} = \frac{(2 \cdot 1 + 2 \cdot 0 + 1 \cdot (-2))}{2^2 + 2^2 + 1^2} (2, 2, 1)$$
$$= \frac{0}{9} (2, 2, 1) = (0, 0, 0)$$

The projection of $\vec{a}$ onto $\vec{b}$ is a single point
because $\vec{a}$ and $\vec{b}$ are orthogonal.

$\vec{w}$ is therefore undefined because the zero vector doesn't
have direction.

b) Without knowing what $\vec{w}$ is, this question is not possible.

Q2

$$\vec{p} = \vec{a} + (\vec{a} \times \vec{b}) \quad \|\vec{a}\| = 1 \quad \vec{b} = (0, 0, 1) \quad \|\vec{b}\| = 1$$

let $\vec{a} = (x, y, k)$

$$\text{then } \vec{a} \times \vec{b} = \begin{bmatrix} x & y & k \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} |y & k| & |x & k| & |x & y| \\ 0 & 1 & 0 & 1 & 0 & 0 \end{bmatrix}$$

$$\vec{a} \times \vec{b} = (y, -x, 0)$$

$$\vec{p} = (y, -x, 0) + (x, y, k)$$

$$\vec{p} = (y+x, y-x, k)$$

let $\vec{a} = (\sin \phi \cos \theta, \sin \phi \sin \theta, \cos \phi)$

$$\therefore \vec{p} = (\sin \phi \cos \theta + \sin \phi \sin \theta, \sin \phi \sin \theta - \sin \phi \cos \theta, \cos \phi)$$

$$\|\vec{p}\|^2 = x^2 + y^2 + z^2$$

$$x^2 = \sin^2 \phi \cos^2 \theta + 2 \sin^2 \phi \sin \theta \cos \theta + \sin^2 \phi \sin^2 \theta$$

$$y^2 = \sin^2 \phi \sin^2 \theta - 2 \sin^2 \phi \sin \theta \cos \theta + \sin^2 \phi \cos^2 \theta$$

$$z^2 = \cos^2 \phi$$

$$x^2 + y^2 + z^2 = 2 \sin^2 \phi \cos^2 \theta + 2 \sin^2 \phi \sin^2 \theta + \cos^2 \phi$$

$$= 2 \sin^2 \phi (\cos^2 \theta + \sin^2 \theta) + \cos^2 \phi$$

$$= 2 \sin^2 \phi (1) + \cos^2 \phi$$

$$= 2(1 - \cos^2 \phi) + \cos^2 \phi$$

$$z^2 = \cos^2 \phi$$

$$\therefore x^2 + y^2 + z^2 = 2(1 - z^2) + z^2 = 2 - z^2$$

$$x^2 + y^2 + 2z^2 = 2$$

$\therefore$ The equation is constant and not it can only move according to ϕ and θ each therefore is on a sphere.

Q3

a) $w = 4\sqrt{2}(-1+i)$

$$|w| = 4\sqrt{2} \cdot (\sqrt{(-1)^2 + 1^2})$$

$$= 4\sqrt{2} \cdot \sqrt{2}$$

$$= 4 \cdot 2$$

$$|w| = 8$$

$$a = |w| \cos \theta$$

$$b = |w| \sin \theta$$

$$-4\sqrt{2} = 8 \cos \theta$$

$$4\sqrt{2} = 8 \sin \theta$$

$$\cos \theta = -\frac{4\sqrt{2}}{8}$$

$$\sin \theta = \frac{4\sqrt{2}}{8}$$

$$\theta = \cos^{-1}(-\frac{1}{2})$$

$$\theta = \sin^{-1}(\frac{1}{2})$$

$$\theta = \frac{3\pi}{4}$$

$$\theta = \frac{\pi}{4}$$

w is in the second quadrant, -1 on the x -axis
 $+i$ in the imaginary axis.

$$\therefore \theta = \frac{3\pi}{4}$$

$$Z = |Z|(\cos \theta + i \sin \theta)$$

$$Z^n = |Z|^n (\cos n\theta + i \sin n\theta)$$

$$n = \frac{1}{3} \quad \theta = \frac{3\pi}{4}$$

$$\therefore Z_k = 8^{\frac{1}{3}} (\cos [\frac{1}{3}(\frac{3\pi}{4} + 2\pi k)] + i \sin [\frac{1}{3}(\frac{3\pi}{4} + 2\pi k)])$$

$$= 2 (\cos [\frac{1}{3}(\frac{3\pi}{4} + 2\pi k)] + i \sin [\frac{1}{3}(\frac{3\pi}{4} + 2\pi k)])$$

$$k=0 \quad \rho_1 = 2 (\cos [\frac{1}{3}(\frac{3\pi}{4})] + i \sin [\frac{1}{3}(\frac{3\pi}{4})])$$

$$\rho_1 = 2 (\cos (\frac{\pi}{4}) + i \sin (\frac{\pi}{4}))$$

$$\rho_1 = 2 (\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2})$$

$$\rho_1 = \underline{\underline{\sqrt{2} + i\sqrt{2}}}$$

$$k=1 \quad \rho_2 = 2 (\cos [\frac{1}{3}(\frac{3\pi}{4} + 2\pi)] + i \sin [\frac{1}{3}(\frac{3\pi}{4} + 2\pi)])$$

$$= 2 (\cos [\frac{1}{3}(\frac{11\pi}{4})] + i \sin [\frac{1}{3}(\frac{11\pi}{4})])$$

$$= 2 (\cos (\frac{11\pi}{12}) + i \sin (\frac{11\pi}{12})) \quad [\frac{11\pi}{12} = \frac{9\pi}{12} + \frac{2\pi}{12} = \frac{3\pi}{4} + \frac{\pi}{6}]$$

$$= 2 (\cos (\frac{3\pi}{4} + \frac{\pi}{6}) + i \sin (\frac{3\pi}{4} + \frac{\pi}{6}))$$

$$= 2 (\cos \frac{3\pi}{4} \cos \frac{\pi}{6} - \sin \frac{3\pi}{4} \sin \frac{\pi}{6} + i (\sin \frac{3\pi}{4} \cos \frac{\pi}{6} + \cos \frac{3\pi}{4} \sin \frac{\pi}{6}))$$

$$= 2 ((-\frac{\sqrt{2}}{2})(\frac{\sqrt{3}}{2}) - (\frac{\sqrt{2}}{2})(\frac{1}{2}) + i ((\frac{\sqrt{2}}{2})(\frac{\pi}{6}) + (-\frac{\sqrt{2}}{2})(\frac{1}{2})))$$

$$= 2 (\frac{-\sqrt{6} - \sqrt{2}}{4} + i \frac{\sqrt{6} - \sqrt{2}}{4})$$

$$\rho_2 = \underline{\underline{\frac{-\sqrt{6} - \sqrt{2}}{2} + i \frac{\sqrt{6} - \sqrt{2}}{2}}}$$

$$\begin{aligned}
 k=2 \quad \rho_3 &= 2 \left(\cos \left[\frac{1}{3} \cdot \left(\frac{3\pi}{4} + 4\pi \right) \right] + i \sin \left[\frac{1}{3} \cdot \left(\frac{3\pi}{4} + 4\pi \right) \right] \right) \\
 &= 2 \left(\cos \left[\frac{1}{3} \cdot \left(\frac{19\pi}{4} \right) \right] + i \sin \left[\frac{1}{3} \cdot \left(\frac{19\pi}{4} \right) \right] \right) \\
 &= 2 \left(\cos \left(\frac{19\pi}{12} \right) + i \sin \left(\frac{19\pi}{12} \right) \right) \quad \left[\frac{19\pi}{12} = \frac{\cancel{12\pi}}{12} + \frac{9\pi}{12} + \frac{2\pi}{12} = \frac{3\pi}{4} + \frac{\pi}{6} \right] \\
 &= 2 \left(\cos \left(\frac{3\pi}{4} + \frac{\pi}{6} \right) + i \sin \left(\frac{3\pi}{4} + \frac{\pi}{6} \right) \right) \\
 &= 2 \left(\cos \left(\frac{3\pi}{4} \right) \cos \left(\frac{\pi}{6} \right) - \sin \left(\frac{3\pi}{4} \right) \sin \left(\frac{\pi}{6} \right) + i \left(\sin \left(\frac{3\pi}{4} \right) \cos \left(\frac{\pi}{6} \right) + \cos \left(\frac{3\pi}{4} \right) \sin \left(\frac{\pi}{6} \right) \right) \right) \\
 &= 2 \left(\frac{\sqrt{6}-\sqrt{2}}{4} - i \frac{\sqrt{6}+\sqrt{2}}{4} \right) \\
 &= \frac{\sqrt{6}-\sqrt{2}}{2} - i \frac{\sqrt{6}+\sqrt{2}}{2}
 \end{aligned}$$

→ b) A parallelogram with 3 known sides

$$\begin{aligned}
 \rho_1 + \rho_2 + \rho_3 &= (\sqrt{2} + i\sqrt{2}) + \left(\frac{\sqrt{6}-\sqrt{2}}{2} + i \frac{\sqrt{6}-\sqrt{2}}{2} \right) + \left(\frac{\sqrt{6}-\sqrt{2}}{2} - i \frac{\sqrt{6}+\sqrt{2}}{2} \right) \\
 &= \left(\sqrt{2} + \frac{\sqrt{6}-\sqrt{2}}{2} + \frac{\sqrt{6}-\sqrt{2}}{2} \right) + i \left(\sqrt{2} + \frac{\sqrt{6}-\sqrt{2}}{2} - \frac{\sqrt{6}+\sqrt{2}}{2} \right) \\
 &= (\sqrt{2} - \sqrt{2}) + i(\sqrt{2} - \frac{2\sqrt{2}}{2}) \\
 &= 0 + i0
 \end{aligned}$$

∴ ρ_1, ρ_2 and ρ_3 form a equilateral triangle
vertices A, B, C are known with D being opposite B

$$\therefore \cancel{D=A+B} \therefore D=A+C-B$$

$$\textcircled{D} \therefore \rho_1 + \rho_2 + \rho_3 = 0 \rightarrow A+C=-B$$

$$\therefore D = -B - B = -2B$$

$$\begin{aligned}
 \rho_4 &= -2(\sqrt{2} + i\sqrt{2}) \\
 &= \underline{\underline{-2\sqrt{2} - i2\sqrt{2}}}
 \end{aligned}$$